

open platform

ONE PC, MANY PLATFORMS

Need a Palm to code for a Palm? Not really. You can code and test applications meant for different platforms on your PC...if you know how

You want to develop an application for a mobile phone, a PDA or perhaps for Linux, but you have only one PC with Windows installed on it. You could write the code in Windows but to test it, the normal process would be to have an assembly line of machines with the relevant operating systems (OSes) and compilers installed. But this obviously increases the cost of development. Even installing multiple OSes on different partitions would entail extra time and effort in switching from one OS to another.

A cost-effective solution, especially for students and rookies, is to set up a multi-platform development unit on a single PC, converting your PC into a Swiss Army knife kind of development platform using certain software and emulators.

Jack of all Trades

In addition to Java, several languages offer Virtual Machines (VMs) for multiple platforms. Programs developed in these languages can run on platforms other than the ones in which they were developed. If you want to develop for Windows, Linux, and Palm, use Python. If Windows and Linux are your target platforms, you can use PHP, Perl, and Rebol. If you want to add a graphical interface to the applications you develop with these languages, there are tools for that as well. You will need Cygwin and either GTK or QT windowing libraries for this.

Starting blocks

The least you would need for such a setup is a PC with at least a 500 MHz processor, 256 MB of RAM and a 20 GB hard disk. Anything less than this and you might not be able to run other applications while developing. After all, we are talking about running a whole bunch of development tools and platforms at the same time, which will hog a sizeable amount of processing power.

It's best to install Windows 2000/NT as the base OS on which you can install all your tools. This platform supports almost all the development tools you'll need. The support factor is especially important when it comes to developing for mobile applications, and most mobile development tools are built to operate on the NT platform. It is also pretty stable and user-friendly.

If you want to develop for Linux using Windows, you can use applications such as VMWare. Also check out for development issues within Linux. Several free development tools are also available for Windows, and unless you are really good at Linux, Windows 2000 will do.

Once these are in place, you can install software that allow for virtual machines to be created or use software development kits and emulators specifically built for multi-platform use.

VMware your PC

VMware Workstation (version 3.0 is the current one) is a desktop application that will help you run different OSes on one PC. It creates virtual machines inside a single computer, which can interact with

each other through a virtual network set up by the software inside your PC. For example, if you have Windows 2000 installed on your machine and also want to install Linux, you have to make your PC a dual boot system with a separate partition for Linux. With VMware Workstation you can install Linux as a virtual machine running at the same time as Windows 2000. It can run several such OSes at once, allowing you to test your applications across a virtual network of different platforms. (See the box, 'Using VMware to set up Guest Operating Systems')

MS DOS, Windows (versions 3.11 to XP), Linux (Red Hat, SuSE, and Mandrake) and FreeBSD are supported as guest operating systems, which means you can run these in VMware Workstation after installing them. You can install other OSes but they may not function properly.

You can also buy VMware Workstation with pre-installed guest OSes, but you have to pay more for these packages. A 30-day evaluation version of VMware Workstation is available from www.vmware.com. You will have to register before downloading it to get the evaluation license number.

Go mobile!

To develop for the mobile platforms you can make do with emulators that can be installed on the desktop. But before you start, you need to have some basic utilities on your PC, all of which are available as free downloads.

To develop for the Palm, the SDK (Software Development Kit) of choice is SDK 3.5, which you can get from the Palm